

AI-Generated Steganographic Malware: Evaluating Detection and Evasion in Modern Anti-Virus Systems

Master of Science in Computer Science at the University of
Johannesburg

Student: Thendo Siphuma

Supervisor: Prof Daniel Du Plessis

Table of Contents

2.1. Introduction	1
2.2. Evolution of Steganography	2
2.2.1. Historical and Traditional Techniques	3
2.2.2. Least Significant Bit (LSB) Substitution	5
2.2.3. AI-Generated Steganography	6
2.3. Limitations of Current Antivirus Solutions	8
2.3.1. Ineffectiveness of Signature-Based Detection	8
2.3.2. Failure Against Concealed Malware	9
2.4. Detection Challenges for Steganographic Malware.....	10
2.4.1. Ineffectiveness of Existing Tools	10
2.4.2. Evasion Techniques	11
2.4.3. Need for Advanced Algorithms.....	11
2.5. YARA: Capabilities and Limitations	12
2.5.1. YARA's Role and Capabilities	12
2.5.2. YARA's Limitations	13
2.5.3. The Need for a Hybrid Approach	14
2.6. The Role of VirusTotal in Testing	14
2.7. Emerging Solutions and Approaches	15
2.7.1. Custom YARA Rules	16
2.7.2. Machine Learning-Based Anomaly Detection	17
2.7.3. Hybrid Detection Models.....	18
2.8. Summary	18
References	20

FIGURES

Figure 1: Classification of spatial domain image steganography techniques..... 4

Figure 2: Architecture of the steganography model..... 7

TABLES

YARA RULE demonstration example.....17

Chapter 2: Literature review

2.1. Introduction

Recent advances in generative Artificial Intelligence (AI) (for example, transformer and diffusion models) have not only improved learning experience in education field (Morales-Chan, et al. 2024) but has enabled the expansion of the attack surface in cybersecurity enabling digital concealments such as AI-Generated Steganography as noted by Varney (2019). Generative AI provides cyber adversaries with minimal programming skills a platform to automate the creation of complex malicious code, facilitate malware generation, and conduct sophisticated cyberattacks with reduced technical effort (Metta, et al. 2024). Recent studies by Motwani, et al. (2024) demonstrate how cyber adversaries can leverage AI to automate data hiding, reduce technical skills required to create strong and undetectable threats.

Steganography derived from the Greek words *steganos* (covered) and *graphia* (writing) refers to the art of hiding information (Kour and Verma 2014) in carriers such as media files in such a way that the presence of the information is unknown (Mandal, et al. 2022). Steganography forms part of a larger part of data hiding techniques which also include watermarking (Rustad et al., 2023). Within Steganography three primary categories are commonly identified - text, audio, and image steganography (Singh 2016). This study will focus on image steganography due to its extensive use in hiding malicious payloads while keeping its visual quality and structural integrity (Saputro, et al. 2024).

Steganography techniques directly undermine the effectiveness of traditional antivirus (AV) systems as shown by Dell’Orco, et al. (2025b) on their test on antivirus against elusive malware. Conventional AVs primarily rely on signature-based detection where at least one byte of the code will be compared to an existing signature of already existing malwares, which are stored in a database known (Chakravarty, et al. 2019). Although some of the modern tools incorporate heuristics, rules and AI analysis their performance remains questionable when confronting malwares concealed in benign-looking files (Hussain Hussain and Raheel Abbas 2025). This gap provides a basis for the need for advanced hybrid mechanisms capable of identifying hidden or AI-generated malware.

This chapter reviews literature from 2000 to 2025 from peer-reviewed journals, industry white papers, and public datasets. It traces the evolution of steganography and its intersection with artificial intelligence, examines the limitations of conventional antivirus solutions, and analyses the detection challenges posed by stegomalware. Finally, it explores emerging hybrid detection frameworks that combine YARA rule-based analysis with machine learning (ML) techniques to improve the detection of AI-generated steganographic malware.

2.2. Evolution of Steganography

Steganography is the science that involves communicating secret data in an appropriate multimedia carrier such as an image (Cheddad, et al. 2010). The practice has a rich history and literature which has rapidly developed over the past decades (Sahu and Sahu 2020). Although its roots go back to ancient times when Demaratus (king of Sparta) and Histiaeus (ruler of Miletus) communicated using hidden messages (Sahu and Sahu 2020), the recent rise in technological advancements have moved the practice from inks to digital pixel manipulations (Evsutin, et al. 2020). Understanding this evolution is crucial as it highlights how technological change has continuously increased the detection challenges of steganography for modern detection systems (Bamanga and Babando 2024). Understanding this evolution is crucial, as each technological phase has introduced new concealment techniques that continuously challenge modern detection systems (Kumar, et al. 2024).

The evolution of Steganography can be looked at through three interrelated perspectives that shows how the discipline has advanced alongside computing technologies. These include carrier modality, capacity–detectability trade-offs, and the degree of automation in embedding which collectively reveal how steganography has evolved from simple, human-crafted concealment methods to complex, intelligent, and data-driven systems capable of deceiving modern forensic detection tools as noted below:

Carrier modality:

This dimension explains the evolution of how information has been carried through steganography over time from ancient methods, such as hiding messages under wax tablets as described by (Sahu and Sahu 2020), to modern techniques that use digital media like images, audio, and video fuelled by the rise of digital technologies. Different advanced frameworks now further allow multiple media simultaneously without extensive neural network training, thereby reducing computational costs and improving adaptability (Song, et al. 2024). This expanded range of carrier modalities has increased the potential attack surface, making detection increasingly complex for cybersecurity systems (Mallick and Nath 2024).

Capacity-detectability trade-offs:

A fundamental challenge throughout the history of steganography is achieving an optimal balance between data-hiding capacity and noticeability (Sahu and Sahu 2020). Increasing the amount of data hidden within a carrier file generally introduces more statistical distortions, making the presence of the hidden message easier to detect (Cheddad, et al. 2010). Early digital methods, such as simple Least Significant Bit (LSB) substitution, could achieve high capacity but were vulnerable to statistical steganalysis attacks that could easily spot these modifications (Sahu and Sahu 2020). Frequency-domain methods such as Discrete Cosine Transform (DCT) and Discrete Wavelet Transform (DWT) improved robustness against image processing and

detection, though at the cost of reduced capacity of data hidden (Cheddad, et al. 2010). Modern AI-based models, particularly those employing Generative Adversarial Networks (GANs), address this trade-off by learning embedding patterns that mimic the natural statistical features of carrier images, thereby increasing capacity while minimizing detectability (Liu, et al. 2020). The transition from rule-based to data-driven embedding signifies a significant enhancement in steganographic strength (Luo, et al. 2024).

Degree of automation in embedding:

Early digital methods required substantial human intervention (Sahu and Sahu 2020) but in recent steganography techniques, it has transitioned from manual operations to fully automated, intelligent systems (Majeed, et al. 2021). The beginning of adaptive steganography introduced partial automation by allowing algorithms to analyse cover images and select regions often textured or noisy areas where data could be concealed with minimal impact (Cheddad, et al. 2010). With the rise of deep learning, complete automation has become possible with models such as Generative Adversarial Networks (GANs) and Convolutional Neural Networks (CNNs) autonomously learning from optimal embedding strategies without predefined rules (Saputro, et al. 2024). For example, a GAN-based system trains a generator network to create stego-images realistic enough to deceive a discriminator, thereby producing highly imperceptible and secure embeddings (Liu, et al. 2020). This high level of automation significantly enhances the stealth and complexity of steganographic malware, posing major challenges for traditional antivirus detection systems (Dell'Orco, et al. 2025b).

2.2.1. Historical and Traditional Techniques

In today's world, steganography continues to play a crucial role in ensuring secure and private communications, particularly in environments subject to surveillance and monitoring (Choudhary and Agarwal 2024). This role has actually advanced from ancient times as Sahu and Sahu (2020) explained Demaratus wrote the warning message to Greece about the forthcoming attack inside the piece of wood, which was placed beneath a wax tablet. Likewise, Histiaeus tattooed a message on his messenger's shaved scalp and then waited for the hair to regrow (Diepenbroek 2023) and In World War II invisible ink and microdots were popular methods of hiding information within letters and photographs (Hammad, et al. 2024).

Digital steganography emerged with the ability to manipulate carrier bits commonly using images and audio files to embed secret payloads without obvious visible change (Kumar, et al. 2024). Digital image data is a prevalent medium for steganography due to its high redundancy, providing ample capacity for embedding images (Abdalwahid, et al. 2024). This study focusses on Image steganography, which can be categorized into two domains:

- Transform domain which hides information by modifying the frequency components of a media file, such as using discrete cosine or wavelet transforms, making it more robust against compression and attacks than spatial domain methods (Alyousuf, et al. 2020) and
- Spatial domain steganography (Mandal, et al. 2022), In spatial domain techniques the secret message is hidden inside the image by applying some manipulation over the different pixels of the image steganography (Swain and Lenka 2014) as shown in the figure below:

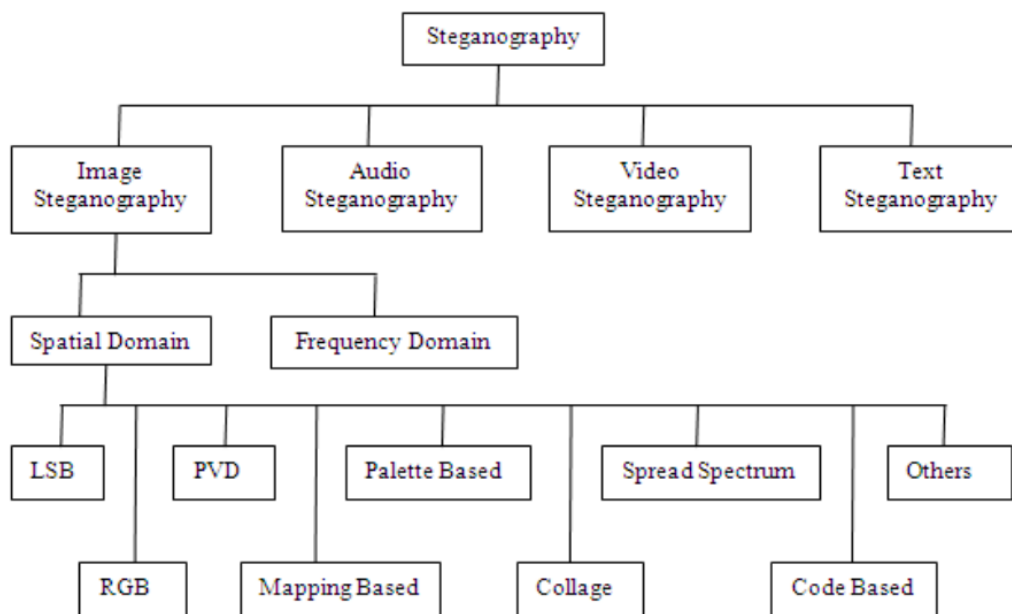


Figure 1: (Swain and Lenka 2014) Classification of spatial domain image steganography techniques

Swain and Lenka (2014) categorize spatial-domain steganography into several techniques, including the, including Least Significant Bit (LSB) method, Pixel Value Differencing (PVD), and RGB-based embedding:

- Least Significant Bit (LSB) : Described by Swain and Lenka (2014), LSB is the simplest and popular image steganographic method is the least significant bit (LSB) substitution and It embeds messages into cover image by replacing the least significant bits directly. To be further under [2.2.2. Least Significant Bit](#).
- Pixel value differencing: Described by Mandal, et al. (2022) conceals secret data by leveraging the fact that edge regions of an image can tolerate greater modification than smooth areas. In this approach, the cover image is divided into non-overlapping pixel pairs, and the secret message is embedded in the pixel-value difference, calculated as $B_i = k_{i+1} - k_i$ where B_i ranges from 0 to 255.

However, this study excludes PVD because it allows for lower data capacity compared to LSB, which distributes information uniformly across all pixels.

- Other spatial-domain techniques, such as RGB-Based hides data in the red, green, and blue channels of pixels using modified LSB approaches to enhance security and minimize visual distortion (Swain and Lenka 2014).

Alongside embedding techniques, steganalysis focuses on detecting, analysing, and potentially extracting hidden information from media to uncover the presence of steganography (Eid, et al. 2022). Steganalysis is often framed using warden models, which define how adversaries interact with hidden communication (Al-Obaidi, et al. 2024). The active warden model proactively disrupts steganography by modifying potential cover objects such as images before they can be used, thereby causing embedding algorithms to fail or produce easily detectable artefacts (Desoky 2011) and passive warden model merely monitors communication channels or files to detect hidden content without altering the transmitted data (Luo, et al. 2024). This study aligns with the passive warden model, as it analyses files non-destructively by examining the bit patterns of the least significant bits (LSBs) across digital images (Aslam, et al. 2022).

2.2.2. Least Significant Bit (LSB) Substitution

The simplest and most popular image steganographic method is the least significant bit (LSB) substitution (Emad, et al. 2018) which embeds messages into cover images by replacing the least significant bits directly (Swain and Lenka 2014). Although this study focuses on LSB embedding in images, the same principle can be applied to other media types such as audio and video, which are also effective steganographic carriers due to their inherent redundancy (Asad, et al. 2011). The simplicity and high embedding capacity of LSB-based approaches make them foundational to modern spatial-domain steganography (Alhomoud 2021).

To reduce detectability and enhance security, several parts of LSB have been developed. One such approach is such as randomized LSB, which modifies pixel insertion positions to avoid predictable sequential patterns (Rafat and Sajjad 2024). ALabaichi, et al. (2020) described how randomization is a common strategy that improves resistance to detection by distributing hidden data irregularly across the image and in relation to this, Rahman, et al. (2025) propose a Magic Matrix approach that shuffles blocks of an image's blue channel before embedding, enhancing resistance to steganalysis. Other methods employ adaptive selection, which targets specific regions of an image, like edges detected via the Canny algorithm, to hide information (Aslam, et al. 2022). Adaptive methods can also vary the number of bits hidden in each pixel to improve the stego-image's quality (Emad, et al. 2018). LSB matching (LSBM) is another widely known variant that alters the embedding logic to be less statistically detectable (ALabaichi, et al. 2020).

Despite its simplicity, Rahman, et al. (2022) showed that the LSB method remains highly vulnerable to statistical steganalysis especially when detection techniques such as histogram analysis and RS (Regular–Singular) testing are introduced as they can expose the subtle distributional shifts introduced by LSB embedding. A high degree of similarity between the histograms indicates minimal distortion, while significant differences can reveal tampering (ALabaichi, et al. 2020). A more powerful and targeted technique is RS steganalysis, a statistical method specifically used to identify the presence of hidden data in LSB-based steganography (Rahman, et al. 2022). Numerous studies confirm that standard LSB substitution methods are susceptible to RS detection and are therefore considered among the easiest steganographic techniques to compromise these limitations highlight the importance of developing adaptive and intelligent variants that can resist statistical and AI-based detection (ALabaichi, et al. 2020).

2.2.3. AI-Generated Steganography

AI-generated steganography uses neural encoders, generative models, or optimization loops to conceal payloads that are difficult to detect inside digital media (Raymond and Rick 2023). This evolution represents a subtle yet potent cyber threat where malicious code is concealed within seemingly innocuous files, rendering traditional cybersecurity defences often powerless (Ali and Bhatti 2024). A notable approach, known as coverless steganography, directly transforms secret data into a photorealistic image rather than embedding it within a pre-existing cover file (Rehman, et al. 2024). Generative models such as Generative Adversarial Networks (GANs) and diffusion models are central to this process, using hidden information to construct the final stego media (Liu, et al. 2020). Within a GAN framework, the competition between the generator and discriminator acts as a loop that continually refines the carrier's realism and stealth, making AI-generated steganographic malware highly resistant to conventional detection (Saputro, et al. 2024).

Fig. 1

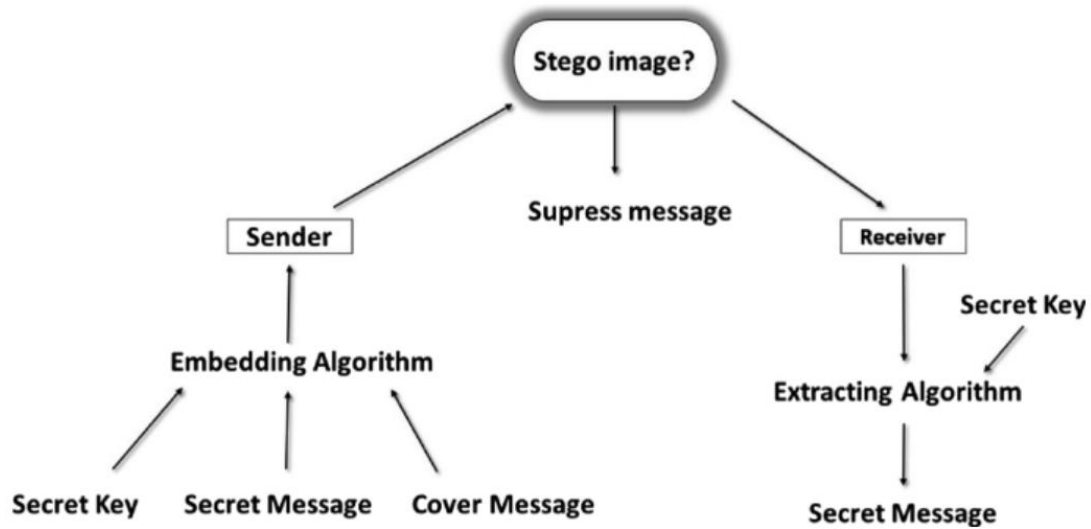


Figure 2 : (Georges and Magdi 2020) Architecture of the steganography model

Encoder-decoder architectures learn mappings between payloads and carrier disquiets, while GANs can synthesize carriers that look natural and embed information during the generating process (Zhang, et al. 2021). In Encoder-decoder models, an encoder can be explicitly trained to embed a secret payload into an image, while a corresponding decoder is trained to extract it, creating a structured method for concealment (Gao, et al. 2022). In contrast, GAN-based concealment produces a stego-image that is visually very natural and difficult to distinguish from an authentic one (Saputro, et al. 2024). This is a significant departure from traditional techniques like Least Significant Bit (LSB) substitution, which operate at the bit or pixel level by modifying an existing image. AI-powered methods instead embed data at a semantic or structural level, making the concealment an integral part of the media's fabric (Raymond and Rick 2023). Because this strategy leaves no conventional artefacts for signature-based tools like YARA to detect, it poses a challenge to modern antivirus systems (Almomani, et al. 2022).

AI enables semantic-level embeddings that preserve contextual coherence and can be optimized against detection models, thereby increasing stealth and cross-detector transferability (Raymond and Rick 2023). **Operating at this semantic level allows malicious data to be integrated within the meaning of the content itself without disrupting its logical or structural consistency.** Chang and Echizen (2025) demonstrate how information can be hidden in the linguistic domain of audiovisual files, preserving semantic equivalence and rendering the concealment resistant to spatial or temporal analysis. Such semantic plausibility makes it difficult for existing machine-learning-based anomaly detectors to recognize statistical irregularities. Moreover, adversarial training further strengthens model robustness against steganalysis tools explained by Gao, et al. (2022) who further note As a result, detection accuracy for some AI-generated stego-images drops to approximately 50 percent which is equivalent to

randomly guessing which highlight the urgent need for more advanced detection frameworks.

Despite their power, AI-driven methods lack standardized benchmarks and raise reproducibility questions, complicating assessments of Anti-Virus resilience (Dell'Orco, et al. 2025b). The primary forensic challenge in detecting AI-generated steganographic malware is that there is often no "original" carrier file for comparison, which undermines traditional detection methods that rely on identifying statistical deviations (Raymond and Rick 2023). This complicates both YARA rule creation and the training of anomaly detection models, which benefit from a clear baseline of what constitutes a "clean" file. This problem is magnified when threat actors use private or fine-tuned AI systems, as it becomes nearly impossible to reverse-engineer the logic to create reliable detection signatures (Raymond and Rick 2023). As a result, AV systems often fail to detect malware concealed through steganographic techniques (Almomani, et al. 2022). The ability for these techniques to carry viruses and malware presents major risks, justifying the need for a hybrid detection pipeline that combines signature-based and anomaly-based approaches to counter this evolving threat (Williams 2019).

2.3. Limitations of Current Antivirus Solutions

As demonstrated in the previous sections, AI-generated steganography integrates malicious data within benign files producing realistic and undetectable files (Williams 2019). These advances challenge traditional antivirus (AV) systems which primarily rely on static signature-based or heuristic scanning (Hussain Hussain and Raheel Abbas 2025) as concealment leaves no recognizable code fragments or metadata anomalies as shown by Dell'Orco, et al. (2025b). This section critically evaluates these limitations, focusing on how conventional detection architectures fail against AI-enhanced steganographic malware and why hybrid, adaptive mechanisms are now required.

2.3.1. Ineffectiveness of Signature-Based Detection

Signature based detection works by looking for certain known byte sequences, code fragments, or digital fingerprints in a large database of signatures which is one of the key process in any virus system (Al-Asli and Ghaleb 2019). These techniques struggle to work when confronted with AI-generated steganography which embeds payloads in ways that leave no detectable signatures (Raymond and Rick 2023). Modern generative models such as Generative Adversarial Networks (GANs) and encoder-decoder architectures produce stego-images that appear entirely authentic by concealing data within the semantic or structural composition of media file as noted by (Saputro, et al. 2024) and (Gao, et al. 2022). Unlike traditional Least Significant Bit substitution methods that alter existing pixel data (Rahman, et al. 2022), these AI-

driven methods transform entire images leaving no byte patterns or modification artefacts that signature-based systems can recognize (Raymond and Rick 2023).

It is possible to preserve semantic equivalence between the cover and stego transcripts reducing the likelihood of detection as malicious content is seamlessly integrated within the context of the host file (Chang and Echizen 2025). This creates a serious problem during detection, since there's usually no original version of the file to compare against and without that baseline, it becomes nearly impossible to verify authenticity using traditional statistical or signature-based methods (Raymond and Rick 2023). As a result, antivirus programs that rely on fixed signatures or rule-based detection often can't spot these hidden payloads. Research shows that when malicious code is hidden inside multimedia files, traditional antivirus tools frequently miss them sometimes failing to detect as much as half of the concealed threats (Almomani, et al. 2022).

2.3.2. Failure Against Concealed Malware

While Williams (2019) highlights that the detection of AI-generated steganography is becoming increasingly difficult for cybersecurity experts, exposing organizations to significant risks, Joyce, et al. (2023) provide the technical basis with their findings showing challenges with some deep malware analysis methods as they are computationally expensive and cannot feasibly scale to the volume of files processed by antivirus engines. In one study, while many antivirus engines detected a ransomware sample in its original format, none could identify it after it was concealed within an HEVC video stream (Almomani, et al. 2022). Similarly, image-based concealment has been shown to bypass detection by 90% against commercial antivirus tools (Dell'Orco, et al. 2025b). This evasion is effective because deep scanning is prone to high rates of false positives, as many legitimate applications perform image manipulation operations that are structurally similar to steganographic decoding as further shown by Dell'Orco, et al. (2025b). Attackers take advantage of this weakness by hiding harmful code inside ordinary-looking files, such as JPEG images, allowing them to slip past security software undetected (Knöchel and Karius 2024) and as a result, steganography tools and the malware hidden within them can easily slip past perimeter defences such as web filters and antivirus systems (Frith 2007).

A primary advantage of stegomalware is its ability to bypass static analysis because the malicious payload is only revealed at runtime (Dell'Orco, et al. 2025b). This creates a critical time-to-detection problem, as experiments show it can take several days for AV engines to adapt their signatures after a new threat is submitted Dell'Orco, et al. (2025b) further explained. This operational window allows the malware to establish a connection to its Command & Control (C&C) servers to download additional malicious code (Knöchel and Karius 2024) and attackers often take advantage of this delay to quietly steal sensitive information, using steganographic methods to move files or

confidential company data out of the organization without being noticed (Frith 2007). Because the malicious activity only commences after the payload is extracted from its benign-looking carrier, behaviour-based detection systems are placed at a disadvantage, reacting to the attack only after the initial infiltration has succeeded (Frith 2007).

2.4. Detection Challenges for Steganographic Malware

Anyone may use the steganography techniques to secretly share the data for legitimate purpose, but adversaries constantly look up new attacking ways and may be used steganography for illegitimate attack purpose (Chaganti, et al. 2021). Spotting this kind of stegomalware is especially difficult for security systems because the hidden data blends in so well that it looks just like the harmless content of the original file (Alazab and Pham 2023). Data can be hidden in many different ways such as making changes to tiny pixels in images to subtle language patterns in AI-generated media making it hard for traditional antivirus and steganalysis tools to detect (Luo, et al. 2024). The wide variety of file types, encodings, and encryption methods greatly expands the search space for hidden data, increasing computational costs and false positives (Sanjalawe, et al. 2025) thereby requiring advanced detection methods that integrate statistical, behavioural, and machine learning analysis (Wang, et al. 2023). This section looks at the main technical and practical challenges that make it difficult to reliably detect this growing steganography threat.

2.4.1. Ineffectiveness of Existing Tools

There have been increasing use of security software like anti-virus programs in the current systems (Knöchel and Karius 2024). Commercial antivirus solutions primarily rely on signature-based detection, identifying malware only after its signature has been analysed and catalogued (Dell'Orco, et al. 2025b). As noted, The signature-based approach detects anomalies by comparing system data to the features of known attacks (Kwon, et al. 2022). Since steganography principle is to hide secret information without being noticed by a third party by modifying redundant data in digital media or protocols, such that the carrier's use attributes are not changed during transmission (Wu, et al. 2021) .

(Suarez-Tangil, et al. 2014) found in their study that the current malware detection static techniques fail to identify malicious components when they are obfuscated or embedded separately from the code e.g. hidden in an image, they further explain how stegomalware specimen the malicious component is revealed at execution time only. Thus, it will not match any signature even if the search includes the asset where it is hidden which modern antivirus system use. A study by (Badhani and Muttoo 2018) evaluated ten commercial anti-malware programs to see if they could detect malicious applications hidden inside images using various techniques, nine of the ten programs were unable to detect any of the hiding techniques, while the most effective one still

failed against malware hidden with encryption or steganography. (Dell'Orco, et al. 2025a) further revealed that antivirus systems are ineffective against stegomalware. Although the proposed solution combines steganalysis with dynamic analysis to successfully detect stegomalware, developing a more general detection mechanism independent of the specific embedding scheme is not trivial.

2.4.2. Evasion Techniques

There are a lot of steganography malware evasion techniques, below are noted by (Ghamizi, et al. 2021) :

- EAST (Evasion Attack Steganography): Regarded as a new steganography and watermarking technique based on multi-label targeted evasion attacks. The key idea of EAST is to encode data as the labels of the image that the evasion attacks produce. Its principle is to encode a message as an adversarial image.
- ADV-EMB: A technique that uses traditional statistical heuristics (UNIWARD) to hide the message then an evasion attack against the detector to update the heuristic to make it less detectable.
- SGAN (Steganographic Generative Adversarial Networks): Introduced SGAN a three-part GAN where the generator is trained simultaneously with a discriminator (to ensure the realism of cover-images) and a steganalyser (i.e. a detector that tries to distinguish clean images and stego images) to ensure elusiveness.
- Hidden: Hidden proposes to add a differentiable noise layer between the encoder and the decoder and train the whole model at once. This layer ensures that the loss reconstruction considers potential perturbations of the images and generates robust stego images

2.4.3. Need for Advanced Algorithms

The demonstrated ineffectiveness of existing signature-based tools explain on [2.4.1](#) coupled with the development of sophisticated, AI-driven evasion techniques shown on [2.4.2](#) creates a significant security gap. Suarez-Tangil, et al. (2014) showed that traditional antivirus solutions are fundamentally ill-equipped to handle the dynamic and subtle nature of modern stegomalware, as they rely on detecting known patterns that steganography is explicitly designed to conceal which was further explained by Badhani and Muttoo (2018).

The evasion methods detailed, such as SGAN and EAST, are not merely hiding data; they are leveraging advanced machine learning concepts like Generative Adversarial Networks and adversarial attacks to actively fool detection systems (Ghamizi, et al.

2021). This represents a paradigm shift where attackers are using intelligent algorithms to create steganographic content that is statistically indistinguishable from legitimate media thus a reactive, signature-based defence is no longer viable.

To counter these threats, there is an urgent need for a new generation of advanced detection algorithms. These algorithms must move beyond simple pattern matching and employ machine learning and deep learning models capable of performing sophisticated steganalysis. Such systems are required to identify the subtle, complex statistical artifacts that even advanced steganographic techniques leave behind. To effectively combat AI-powered attacks, the defence must also be AI-powered, capable of adapting to and identifying novel embedding methods as they emerge.

2.5. YARA: Capabilities and Limitations

YARA is a powerful tool for pattern matching and rule-based classification that has become a critical component in modern malware identification and mitigation strategies (Vinay Patil 2025). It enables security professionals to create custom rules that detect specific patterns in files, memory, or network traffic, moving beyond the limitations of traditional hash-based indicators (NCCIC 2015). Understanding both the extensive capabilities and the inherent limitations of this signature-based approach is essential for using it effectively and for developing a comprehensive, layered defence against evolving cyber threats (Vinay Patil 2025).

2.5.1. YARA's Role and Capabilities

YARA, often described as "The pattern matching Swiss knife for malware researchers," serves as a de-facto standard for malware identification and classification (VirusTotal 2019). Its primary function is to identify and classify malware by creating rules that search for specific patterns, which can be either textual or binary in nature (Lockett 2021). Unlike traditional cryptographic hashing that looks for exact file matches, YARA excels at identifying entire malware families by exploiting common code similarities (NCCIC 2015). Coscia, et al. (2025) explained how YARA rule is composed of a meta section for descriptive information, a strings section where patterns are defined, and a condition section that contains a Boolean expression determining when the rule should trigger a match as further explained by Dittmann, et al. (2024).

Key capabilities of YARA include:

- **Flexibility and Customization:** YARA is widely accepted for its flexible and customizable nature, which allows malware analysts to develop rules tailored to the requirements of a specific security domain (Naik, et al. 2020). Rules can

be crafted with great precision using text strings, hexadecimal strings, and regular expressions to define patterns (Coscia, et al. 2025).

- **Platform Support and Integration:** It is an open-source, multi-platform tool that works on Windows, Linux, and Mac OS explained by Naik, et al. (2020) and further noted that YARA can be integrated as a library into other applications and programming languages, including popular antivirus engines like ClamAV, Python, and C/C++.
- **Versatility in Analysis:** The tool can be used for both static and dynamic malware analysis (Naik, et al. 2020). It is effective at detecting obfuscated malware, as rules can be built to identify specific obfuscation techniques like packing or anti-debugging capabilities (Lockett 2021).
- **Extensibility:** For conditions that are too complex for the default syntax, YARA's functionality can be extended with custom modules that provide additional metrics and functions not available in the default toolset (Dittmann, et al. 2024).

2.5.2. YARA's Limitations

Despite its widespread use and capabilities, YARA is not a complete solution and has several limitations, many of which are inherent to its nature as a signature-based detection tool (Lockett 2021).

- **Evasion by Attackers:** The effectiveness of YARA rules is dependent on their defined Indicator of Compromise (IoC) strings, but attackers can manipulate, encrypt, or replace these strings to evade detection (Naik, et al. 2020). Static rules that rely on fixed byte sequences are often ineffective against polymorphic or metamorphic malware, which is designed to constantly change its code structure to avoid signature matching (Vinay Patil 2025).
- **Rule Creation and Maintenance:** Writing high-quality, effective YARA rules is a tedious process that requires a highly specialized skill-set and expert knowledge of malware analysis. As new malware variants are constantly developed, the continuous maintenance, refinement, and updating of rule sets is a complex and time-consuming task (Khalid, et al. 2020).
- **Balancing False Positives and Negatives:** Crafting a robust rule involves a difficult trade-off between sensitivity and precision. A rule that is too specific might miss new malware variants, while a rule that is too general can produce a high number of false positives by incorrectly flagging benign files (Vinay Patil 2025). The success of YARA depends on achieving a challenging balance of the right types and number of IoC strings (Naik, et al. 2020).

- **Detection of Novel Threats:** YARA is most effective in detecting malware that is like existing families and variants. It may fail to detect entirely new or unique malware strains for which no signatures have yet been created (Naik, et al. 2020).
- **Automated Rule Generation:** While tools that automatically generate YARA rules exist, their output often requires manual post-processing and optimization to be effective and may not perform as well as rules created manually by an expert analyst (Naik, et al. 2020).

2.5.3. The Need for a Hybrid Approach

The limitations of YARA as a standalone, signature-based tool highlight the necessity of adopting a hybrid strategy that combines YARA with other detection techniques to create a more resilient and effective defence (Vinay Patil 2025). Since static YARA rules can be bypassed, especially by evasive malware, they must be augmented with more dynamic and flexible methods (Vinay Patil 2025).

One proposed hybrid solution is to enhance YARA rules by integrating them with fuzzy hashing (Naik, et al. 2020). Fuzzy hashing complements YARA by identifying structural similarities between files, which is effective even when specific strings or byte sequences have been modified (Naik, et al. 2020). This combined search can increase the accuracy of detection and the overall confidence in the analysis results as (Naik, et al. 2020) explained and further research has also suggested building fuzzy hashing functionality directly into YARA rules to strengthen them (Lockett 2021).

Furthermore, the weakness of signature-based detection against new and evasive threats highlights the need to incorporate behaviour-based analysis (Lockett 2021). A hybrid strategy that combines YARA's static pattern matching with dynamic analysis which focuses on runtime actions such as suspicious API calls, system modifications, and network activity is far more effective at spotting complex risks (Vinay Patil 2025). Finally, research is exploring the integration of machine learning with YARA to automatically generate more adaptive and intelligent rules, helping to overcome the difficulties and time constraints of manual rule creation (Vinay Patil 2025).

2.6. The Role of VirusTotal in Testing

VirusTotal is a free online tool that one of the most popular malware scanning platforms (Leka, et al. 2022) which plays a crucial role in the testing and analysis of potentially malicious files and URLs (Masri and Aldwairi 2017). Its primary function is to analyse submissions using a vast collection of different antivirus engines, website scanners, and domain blacklists to detect various types of malwares, including viruses, worms, and trojans (Masri and Aldwairi 2017). Developed by VirusTotal, the it contains the YARA framework as act as a testament to the organization's central role in the malware research community (Dittmann, et al. 2024). In experimental and research settings,

VirusTotal is frequently used as a definitive source to validate whether samples are malicious and to verify their correct classifications. For example, one can script the upload of a URL or binary and retrieve the “detection ratio” (number of engines that label it malicious) (Lockett 2021). Its extensive database of known malware hashes and domains makes it invaluable for validation: researchers frequently cross-reference their samples against VirusTotal to ensure correct labelling (Masri and Aldwairi 2017).

Beyond manual uploads, and apart from its primary public interface, PC uploaders, browser extensions, and an Application Programming Interface (API) are all viable avenues for the transmission of data to VirusTotal (Vasani, et al. 2023). A system can be designed to automatically submit URLs to VirusTotal and parse the detection ratio to classify them as malicious or benign (Masri and Aldwairi 2017). Its extensive and reliable threat database contains more than 70 anti-malware analysers and URL/domain blacklisting services in its inspections, in addition to other methods for discriminating against dubious matters (Vasani, et al. 2023). However, while indispensable for testing, its performance is not infallible as Leka, et al. (2022) explained there is no clear understanding of how this platform delivers its results and since the platform operates basically as a Blackbox, the produced results may be questionable. In one study evaluating its effectiveness in detecting malicious advertisements by Masri and Aldwairi (2017) VirusTotal achieved a testing accuracy of 56%, which, while significant, was outperformed by other tools such as URL void with 73% that incorporated a wider range of domain blacklists in addition to scanning engines and highlight that while VirusTotal is a powerful and widely used testing platform, its effectiveness can be enhanced when used as part of a broader, multi-tool testing strategy.

The selection of VirusTotal as the primary validation tool for this study is justified by it being a free and accessible online service, it presents a simple yet powerful solution for large-scale analysis, eliminating the complexities and costs associated with acquiring and maintaining dozens of individual antivirus products. This study will leverage its Application Programming Interface (API) to systematically upload AI-generated steganographic samples containing hidden payloads. The resulting "detection ratio" metric indicating how many of the 70+ integrated, industry-leading antivirus engines flag a sample as malicious will serve as the core measure of evasion success. A low detection ratio will provide powerful, empirical evidence of the AI steganography's ability to bypass the current threat detection landscape. Therefore, Virus Total's comprehensive and standardized approach provides a credible, large-scale benchmark that is essential for strengthening the experimental validation component of this research and solidifying its conclusions on vulnerabilities within modern antivirus systems.

2.7. Emerging Solutions and Approaches

Using steganography during an attack is relatively easy but protecting against it is much more complicated as threat actors are getting more innovative and more creative

(Kaspersky 2023). The core challenge lies in the fact that steganography's primary goal is to hide the very existence of a secret message, making it difficult to detect and even more challenging to understand but they do leave behind detectable traces in the cover medium (Provos and Honeyman 2003).

Traditionally, antivirus systems use signature detection, which relies on a vendor-generated database (Hussain Hussain and Raheel Abbas 2025). This method is only effective against known threats. With the rapid evolution of malware, particularly zero-day attacks, signature-based detection often falls short in identifying new or mutated threats (Hussain Hussain and Raheel Abbas 2025). This steganography vulnerability has necessitated the development of more sophisticated and multi-layered defence frameworks (Prathyusha, et al. 2025) and some emerging solutions are moving beyond simple signature matching to embrace more dynamic and intelligent methods of analysis, focusing on three key areas such as:

- The targeted pattern-matching of custom YARA rules,
- The statistical anomaly detection of machine learning, and
- The synergistic power of hybrid models.

2.7.1. Custom YARA Rules

YARA is a tool aimed at (but not limited to) helping malware researchers to identify and classify malware samples (Ningappa 2020). It functions by creating custom rules that match specific, predefined patterns within a file's binary or text content (Vinay Patil 2025). This signature-based detection approach is a fundamental technique used by antivirus and malware analysis software (Lockett 2021). The value of custom rules lies in their ability to be tailored to the unique characteristics of specific threats, such as the subtle artifacts left by steganography (Khalid, et al. 2020). While YARA is a widely accepted and flexible tool for identifying known patterns, its effectiveness is limited against polymorphic malware, which is designed to constantly change its signature to evade detection (Vinay Patil 2025). Despite this limitation, custom YARA rules remain an essential and efficient tool for threat hunting and providing a rapid method for identifying known malicious indicators (NCCIC 2015).

The study will follow the process described by Naik, et al. (2020) who explained how YARA at its core is designed to spot malware by matching its characteristics, or "signatures," against a database of known threats. They further explain how it works by scanning files and processes for these tell-tale signs and is built with three main parts which are the meta, strings, and condition.

Below is an example of a simple YARA Rule syntax to be employed in the study (Naik, et al. 2021):

```

rule RuleName {

    meta:
        description = "description of rule"
        author = "name"
        date = "dd/mm/yyyy"
        reference = "url"

    strings:
        $text_string1 = "text1 you wish to find in malware"
        $text_string2 = "text2 you wish to find in malware"
        $hex_string1 =
        $hex_string2 =
        $reg_exp_string1 = /regular expression1 you wish to find in malware/
        $reg_exp_string2 = /regular expression2 you wish to find in malware/

    condition:
        $text_string1 or $text_string2 or
        $hex_string1 or $hex_string2 or
        $reg_exp_string1 or $reg_exp_string2
}

```

(Regeciova 2022) explains the different sections involved in detail from the strings section is where patterns been hunted are defined which can further fall into three categories :

1. **Text Strings:** Simple, readable text. You can add modifiers to make the search case-insensitive (no case) or specify the character type (ascii, wide).
2. **Hexadecimal Strings:** Used for matching specific sequences of raw bytes, which is great for finding non-textual data. This format is flexible, allowing for wildcards and other advanced patterns.
3. **Regular Expressions:** A very powerful option, available since version 2.0, that allows you to define complex and variable patterns.

The final piece they explained is the condition section. This is the rule's brain, using a logical statement e.g. if 2 of the 3 strings are found, then it's a match to decide whether to officially flag the file as malware. It's this condition that ultimately determines if the rule triggers.

2.7.2. Machine Learning-Based Anomaly Detection

To overcome the limitations of static signatures, emerging solutions are increasingly turning to machine learning (ML) (Adhav, et al. 2024). Machine learning gives

computers the ability to learn from data without being explicitly programmed, allowing them to discern complex patterns that traditional methods miss and by utilizing unsupervised learning, a system can analyse unlabelled data to discover hidden patterns and classify outliers on its own (Naeem, et al. 2023). ML-based anomaly detection identifies threats by detecting behaviours and observations that deviate significantly from a baseline of normal activity, rather than looking for a known signature (Zehra, et al. 2023). This approach is particularly effective against unknown or "zero-day" threats because it does not rely on prior knowledge of a specific attack. This makes ML-based anomaly detection a powerful method for flagging suspicious files based on statistical irregularities and unusual behaviours that might indicate concealed malware (Stoian 2020).

Machine learning techniques are based on explicit or implicit model that enables the patterns analysed to be categorized into Genetic Algorithms, Fuzzy Logic, Neural Networks, Bayesian networks and outlier detection:

2.7.3. Hybrid Detection Models

Recognizing the individual weaknesses of signature-based and purely behavioural systems, the most robust emerging solution is the development of hybrid detection models (Vinay Patil 2025). A hybrid approach combines the speed and efficiency of signature-based tools like YARA for known threats with the adaptability of machine learning for unknown and evasive malware (Adhav, et al. 2024). This layered strategy creates a more comprehensive defence, using different techniques to cover a wider range of attack vectors (Vinay Patil 2025) and Such an integrated framework, combining static analysis with dynamic or ML-based methods, is designed to improve overall detection accuracy while simultaneously reducing the rate of false positives. By leveraging the complementary strengths of each method, a hybrid model can provide a more resilient and effective defence against the evolving threat landscape (Naik, et al. 2020).

2.8. Summary

The literature review systematically examines the existing body of knowledge on steganography and malware detection to provide the foundational context for this study. The review traces the evolution of steganography, highlighting its progression from historical practices to modern, AI-driven digital concealment techniques that present significant detection challenges. A central theme identified is the significant limitation of current antivirus solutions, which are often based on conventional signature-based detection and frequently fail to identify malware once it is concealed within multimedia files. This critical detection gap is substantiated by studies

demonstrating that even known malware, such as ransomware, becomes invisible to dozens of antivirus engines after being embedded using steganographic techniques.

The review also assesses the capabilities and inherent limitations of YARA, a key tool for pattern-based malware detection. While YARA offers flexibility in developing custom rules, its effectiveness as a signature-based tool is inconsistent against the evasive and evolving nature of steganographic threats.

Finally, the review explores emerging solutions that aim to overcome these challenges. There is a clear trend towards more adaptive systems, with a strong emphasis on machine learning-based anomaly detection as a promising approach for identifying concealed threats through statistical irregularities rather than known signatures. The findings confirm a distinct gap in current detection methodologies and strongly support the rationale for this study's proposed hybrid model, which integrates enhanced YARA rules with machine learning to counter the threat of AI-generated steganographic malware.

References

Abdalwahid, Shadan MJa, et al.

2024 Investigating the Effectiveness of Artificial Intelligence in Watermarking and Steganography for Digital Media Security. 2024 21st International Multi-Conference on Systems, Signals & Devices (SSD), 2024, pp. 552-561. IEEE.

Adhav, Savita, et al.

2024 Tackling the Emerging Threat of Image-Based Steganographic Malware with Advanced Machine Learning Techniques. 2024 5th International Conference on Data Intelligence and Cognitive Informatics (ICDICI), 2024, pp. 74-79. IEEE.

Al-Asli, Mohammed, and Taher Ahmed Ghaleb

2019 Review of signature-based techniques in antivirus products. 2019 International conference on computer and information sciences (ICCIS), 2019, pp. 1-6. IEEE.

Al-Obaidi, Sumia Abdulhussien Razooqi, Mina Zolfy Lighvan, and Mohammad Asadpour

2024 Enhanced image steganalysis through reinforcement learning and generative adversarial networks. Intelligent Decision Technologies 18(2):1077-1100.

ALabaichi, Ashwak, Maisa'a Abid Ali K Al-Dabbas, and Adnan Salih

2020 Image steganography using least significant bit and secret map techniques. International journal of electrical & computer engineering (2088-8708) 10(1).

Alazab, Mamoun, and Tuan Pham

2023 Stegomalware: A Systematic Survey of Malware Hiding and Detection in Images, Machine Learning Models and Research Challenges. Authorea Preprints.

Alhomoud, Adeeb M

2021 Image steganography in spatial domain: Current status, techniques, and trends. Intelligent Automation & Soft Computing 27(1).

Ali, Atif, and Baber Majid Bhatti

2024 Spies in the Bits and Bytes: The Art of Cyber Threat Intelligence: CRC Press.

Almomani, Iman, Aala AlKhayer, and Walid El-Shafai

2022 Novel Ransomware Hiding Model Using HEVC Steganography Approach. Computers, Materials & Continua 70(1).

Alyousuf, Farah Qasim Ahmed, Roshidi Din, and Alaa Jabbar Qasim

2020 Analysis review on spatial and transform domain technique in digital steganography. Bulletin of Electrical Engineering and Informatics 9(2):573-581.

Asad, Muhammad, Junaid Gilani, and Adnan Khalid

- 2011 An enhanced least significant bit modification technique for audio steganography. International Conference on Computer Networks and Information Technology, 2011, pp. 143-147. IEEE.
Aslam, M. A., et al.
- 2022 Image Steganography using Least Significant Bit (LSB) - A Systematic Literature Review. 2022 2nd International Conference on Computing and Information Technology (ICCIIT), 2022, pp. 32-38.
Badhani, Shikha, and Sunil K Mutttoo
- 2018 Evading android anti-malware by hiding malicious application inside images. International Journal of System Assurance Engineering and Management 9(2):482-493.
Bamanga, Mahmud Ahmad, and Aliyu Kamalu Babando
- 2024 Recent Advances in Steganography. Steganography-The Art of Hiding Information: The Art of Hiding Information:23.
Chaganti, Rajasekhar, et al.
- 2021 Stegomalware: A systematic survey of MalwareHiding and detection in images, machine LearningModels and research challenges. arXiv preprint arXiv:2110.02504.
Chakravarty, Adit Kumar, et al.
- 2019 A study of signature-based and behaviour-based malware detection approaches. Int. J. Adv. Res. Ideas Innov. Technol 5(3):1509-1511.
Chang, Ching-Chun, and Isao Echizen
- 2025 Steganography Beyond Space-Time With Chain of Multimodal AI Agents. arXiv:2502.18547.
Cheddad, Abbas, et al.
- 2010 Digital image steganography: Survey and analysis of current methods. Signal processing 90(3):727-752.
Choudhary, Utkarsh, and Parul Agarwal
- 2024 Image Steganography Combined with Cryptography for Covert Communication. Proceedings of the 2024 Sixteenth International Conference on Contemporary Computing, 2024, pp. 207-212.
Coscia, Antonio, et al.
- 2025 APIARY: An API-based automatic rule generator for yara to enhance malware detection. Computers & Security 153:104397.
Dell'Orco, Danilo, et al.
- 2025a Would you mind hiding my malware? Building malicious Android apps with StegoPack. Pervasive and Mobile Computing:102060.
Dell'Orco, Danilo, et al.
- 2025b Challenging Antivirus against Elusive Android Malware over Time. Proceedings of the 2025 Italian Conference on Cybersecurity, ITASEC. CEURWS. org, 2025b.
Desoky, Abdelrahman

- 2011 Edustega: an education-centric steganography methodology. International Journal of Security and Networks 6(2-3):153-173.
Diepenbroek, Martine
- 2023 Secretive Spartans: Herodotus' views towards secret communication in Persia and Sparta and its effect on post-Herodotean sources.
Dittmann, Jana, et al.
- 2024 Forensic trace analysis for MP3 based stego-malware: Exemplary study for stego-algorithm and capacity attribution to derive YARA rules for malware identification. Proceedings of the 2024 ACM Workshop on Information Hiding and Multimedia Security, 2024, pp. 101-112.
Eid, Wafa M, et al.
- 2022 Digital image steganalysis: current methodologies and future challenges. Ieee Access 10:92321-92336.
Emad, Elshazly, et al.
- 2018 A secure image steganography algorithm based on least significant bit and integer wavelet transform. Journal of Systems Engineering and Electronics 29(3):639-649.
Evsutin, Oleg, Anna Melman, and Roman Meshcheryakov
- 2020 Digital steganography and watermarking for digital images: A review of current research directions. IEEE Access 8:166589-166611.
Frith, David
- 2007 Steganography approaches, options, and implications. Network Security 2007(8):4-7.
Gao, Kai, et al.
- 2022 Steganographic secret sharing via AI-generated photorealistic images. EURASIP Journal on Wireless Communications and Networking 2022(1):119.
Georges, Jeanne, and Dalia A Magdi
- 2020 Using artificial intelligence approaches for image steganography: a review. Internet of Things—Applications and Future: Proceedings of ITAF 2019:239-247.
Ghamizi, Salah, et al.
- 2021 Evasion attack steganography: Turning vulnerability of machine learning to adversarial attacks into a real-world application. Proceedings of the IEEE/CVF International conference on computer vision, 2021, pp. 31-40.
Hammad, Omar Mejbek, et al.
- 2024 An Overview of Image Steganography Techniques: Historical Development, Methodologies, and Evaluation Criteria. SHIFRA 2024:74-87.
Hussain Hussain, and Raheel Abbas
- 2025 Evaluating antivirus software: a comparative analysis of detection methodologies and performance metrics in modern antivirus solutions.
Joyce, Robert J, et al.

2023 Avscan2vec: Feature learning on antivirus scan data for production-scale malware corpora. Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security, 2023, pp. 185-196.

Kaspersky

2023 What is steganography? Definition and explanation.

Khalid, Myra, et al.

2020 Automatic YARA rule generation. 2020 International Conference on Cyber Warfare and Security (ICCWS), 2020, pp. 1-5. IEEE.

Knöchel, Mandy, and Sebastian Karius

2024 Text Steganography Methods and their Influence in Malware: A Comprehensive Overview and Evaluation. Proceedings of the 2024 ACM Workshop on Information Hiding and Multimedia Security, 2024, pp. 113-124.

Kour, Jasleen, and Deepankar Verma

2014 Steganography techniques—A review paper. International Journal of Emerging Research in Management & Technology ISSN:2278-9359.

Kumar, Arvind, et al.

2024 Steganography: Unveiling techniques and research agenda.

Kwon, Hee-Yong, Taesic Kim, and Mun-Kyu Lee

2022 Advanced intrusion detection combining signature-based and behavior-based detection methods. Electronics 11(6):867.

Leka, Christian, et al.

2022 A comparative analysis of virustotal and desktop antivirus detection capabilities. 2022 13th International Conference on Information, Intelligence, Systems & Applications (IISA), 2022, pp. 1-6. IEEE.

Liu, Jia, et al.

2020 Recent advances of image steganography with generative adversarial networks. IEEE Access 8:60575-60597.

Lockett, Adam

2021 Assessing the effectiveness of yara rules for signature-based malware detection and classification. arXiv preprint arXiv:2111.13910.

Luo, Weiqi, et al.

2024 A Comprehensive Survey of Digital Image Steganography and Steganalysis. APSIPA Transactions on Signal and Information Processing 13(1).

Majeed, Mohammed Abdul, et al.

2021 A review on text steganography techniques. Mathematics 9(21):2829.

Mallick, Md Abu Imran, and Rishab Nath

2024 Navigating the cyber security landscape: A comprehensive review of cyber-attacks, emerging trends, and recent developments. World Scientific News 190(1):1-69.

Mandal, Pratap Chandra, et al.

2022 Digital image steganography: A literature survey. Information sciences 609:1451-1488.

Masri, Rima, and Monther Aldwairi

2017 Automated malicious advertisement detection using virustotal, urlvoid, and trendmicro. 2017 8th International Conference on Information and Communication Systems (ICICS), 2017, pp. 336-341. IEEE.

Metta, Shivani, et al.

2024 Generative AI in cybersecurity. arXiv preprint arXiv:2405.01674.

Morales-Chan, Miguel, Hector R Amado-Salvatierra, and Rocael Hernandez-Rizzardini

2024 AI-driven content creation: Revolutionizing educational materials. Proceedings of the Eleventh ACM Conference on Learning@ Scale, 2024, pp. 556-558.

Motwani, Sumeet, et al.

2024 Secret collusion among AI agents: Multi-agent deception via steganography. Advances in Neural Information Processing Systems 37:73439-73486.

Naeem, Samreen, et al.

2023 An unsupervised machine learning algorithms: Comprehensive review. International Journal of Computing and Digital Systems.

Naik, Nitin, et al.

2020 Evaluating automatically generated YARA rules and enhancing their effectiveness. 2020 IEEE Symposium Series on Computational Intelligence (SSCI), 2020, pp. 1146-1153. IEEE.

Naik, Nitin, et al.

2021 Embedded YARA rules: strengthening YARA rules utilising fuzzy hashing and fuzzy rules for malware analysis. Complex & Intelligent Systems 7(2):687-702.

NCCIC

2015 USING YARA FOR MALWARE DETECTION.1.

Ningappa, Siddesh

2020 YARA Based Defence Mechanism Against NFC Based Attacks For Android Devices, Dublin, National College of Ireland.

Prathyusha, Kema, J Vinoj, and KB Manikandan

2025 Exploring Artificial Intelligence Security: A Comparative Study of Adversarial Attacks and Steganographic Defenses. 2025 4th International Conference on Sentiment Analysis and Deep Learning (ICSADL), 2025, pp. 20-25. IEEE.

Provos, Niels, and Peter Honeyman

2003 Hide and seek: An introduction to steganography. IEEE security & privacy 1(3):32-44.

Rafat, Khan Farhan, and Syed Muhammad Sajjad

2024 Advancing reversible LSB steganography: Addressing imperfections and embracing pioneering techniques for enhanced security. IEEE Access.

Rahman, Shahid, et al.

- 2025 A novel and efficient digital image steganography technique using least significant bit substitution. Scientific Reports 15(1):107.
Rahman, Shahid, et al.
- 2022 A novel steganography technique for digital images using the least significant bit substitution method. IEEE Access 10:124053-124075.
Raymond, Jessica, and Lopez Rick
- 2023 AI-POWERED STEGANOGRAPHY: HIDDEN COMMUNICATIONS IN GENERATIVE MEDIA ON THE DARKNET.
Regeciova, Dominika
- 2022 Yara: Down the Rabbit Hole Without Slowing Down. The Journal on Cybercrime and Digital Investigations 7(1):17-21.
Rehman, Hafiz Abdul, et al.
- 2024 Leveraging coverless image steganography to hide secret information by generating anime characters using GAN. Expert Systems with Applications 248:123420.
Sahu, Aditya Kumar, and Monalisa Sahu
- 2020 Digital image steganography and steganalysis: A journey of the past three decades. Open Computer Science 10(1):296-342.
Sanjalawe, Yousef, et al.
- 2025 A deep learning-driven multi-layered steganographic approach for enhanced data security. Scientific Reports 15(1):4761.
Saputro, Indrawan Ady, et al.
- 2024 AI-Powered Steganographic Techniques: A Comparison of Traditional Methods and Modern Machine Learning Approaches. 2024 6th International Conference on Cybernetics and Intelligent System (ICORIS), 2024, pp. 01-06. IEEE.
Singh, Harjit
- 2016 Analysis of different types of steganography. International journal of scientific research in science, engineering and technology 2:578-582.
Song, Sojeong, et al.
- 2024 Implicit Steganography Beyond the Constraints of Modality. European Conference on Computer Vision, 2024, pp. 289-304. Springer.
Stoian, Nicolas-Alin
- 2020 Machine learning for anomaly detection in iot networks: Malware analysis on the iot-23 data set, University of Twente.
Suarez-Tangil, Guillermo, Juan E Tapiador, and Pedro Peris-Lopez
- 2014 Stegomalware: Playing hide and seek with malicious components in smartphone apps. International conference on information security and cryptology, 2014, pp. 496-515. Springer.
Swain, Gandharba, and Saroj Kumar Lenka
- 2014 Classification of image steganography techniques in spatial domain: a study. Journal of Computer Science & Engineering Technology (IJCSET) 5(03):219-232.

Varney, Adam

2019 Analysis of the impact of artificial intelligence to cybersecurity and protected digital ecosystems, Utica College.

Vasani, Vatsal, et al.

2023 Comprehensive analysis of advanced techniques and vital tools for detecting malware intrusion. Electronics 12(20):4299.

Vinay Patil, Naveen Kumar M, Pawan Singh M, Ashishika Singh

2025 Effectively Writing YARA Rules to Detect Malware.

VirusTotal

2019 YARA in a nutshell.

Wang, Zihan, et al.

2023 Data hiding with deep learning: A survey unifying digital watermarking and steganography. IEEE Transactions on Computational Social Systems 10(6):2985-2999.

Williams, Octavia C

2019 What are the cybersecurity risks of artificial intelligence generated steganography?, Utica College.

Wu, Zhijun, et al.

2021 Steganography and steganalysis in voice over IP: A review. Sensors 21(4):1032.

Zehra, Sehar, et al.

2023 Machine learning-based anomaly detection in NFV: A comprehensive survey. Sensors 23(11):5340.

Zhang, Chaoning, et al.

2021 A brief survey on deep learning based data hiding. arXiv preprint arXiv:2103.01607.